

GRADE 7 ACCELERATED · UNIT 3

Exponents and Scientific Notation

Ten lessons in teaching order · the next slide jumps to each one

Contents

Click a lesson to jump to it. Each lesson numbers its slides from 1, as its Teacher Edition does.

3.01	Product Laws of Exponents	<i>slide 3</i>
3.02	Quotient Laws of Exponents	<i>slide 38</i>
3.03	Exponential Expressions	<i>slide 72</i>
3.04	Negative Exponent Law	<i>slide 106</i>
3.05	Applying Exponent Laws	<i>slide 140</i>
3.06–07	Evaluating and Equivalent Expressions	<i>slide 174</i>
T-A1	Exponent Laws with Variable Bases	<i>slide 209</i>
T-A2	Negative Exponents with Variable Bases	<i>slide 243</i>
3.08	Writing Large Numbers in Scientific Notation	<i>slide 277</i>
3.09	Writing Small Numbers in Scientific Notation	<i>slide 311</i>

Product Laws of Exponents

MA.7.NSO.1.1

I can use the product of powers, power of a product, power of a power and identity laws to rewrite and evaluate a numerical expression.

Last unit you took roots apart.

Today you put powers together — and the base has to match.

Warm-Up

On your own. Four minutes. No notes.

1. $\sqrt{81}$

2. Between which two whole numbers is $\sqrt{40}$?

3. Solve $3x - 7 = 11$

4. Evaluate $\left(\frac{2}{3}\right)^2$

Warm-Up

Answers.

1. $\sqrt{81}$ **9**

2. Between which two whole numbers is $\sqrt{40}$? **6 and 7**

3. Solve $3x - 7 = 11$ **$x = 6$**

4. Evaluate $\left(\frac{2}{3}\right)^2$ **$\frac{4}{9}$**

Notes

Copy the three words.

I. A power is a shorthand for repeated multiplication

A. **base** — the number used as a factor

B. **exponent** — how many times the base is a factor

C. **power** — the whole thing, base and exponent together

$$2^3 = 2 \cdot 2 \cdot 2 = 8 \qquad \left(\frac{2}{3}\right)^2 = \frac{2}{3} \cdot \frac{2}{3} = \frac{4}{9}$$

Notes

Where the rule comes from.

II. Product of powers — same base, add the exponents

$$3^2 \cdot 3^3 = (3 \cdot 3)(3 \cdot 3 \cdot 3) = 3^5$$

$$3^2 \cdot 2^3 = (3 \cdot 3)(2 \cdot 2 \cdot 2) \quad \text{— different bases, no shortcut}$$

Product of powers: $a^m \cdot a^n = a^{m+n}$

Check with values: $3^2 \cdot 3^3 = 9 \cdot 27 = 243 = 3^5$

Notes

III. Power of a product — the exponent goes to every factor

$$(2 \cdot 5)^3 = (2 \cdot 5)(2 \cdot 5)(2 \cdot 5) = (2 \cdot 2 \cdot 2)(5 \cdot 5 \cdot 5) = 2^3 \cdot 5^3$$

Power of a product: $(ab)^m = a^m \cdot b^m$

Works for a product only, never for a sum:

$$(10 + (-2))^3 = 8^3 = 512 \quad \text{but} \quad 10^3 + (-2)^3 = 992$$

Notes

IV. Power of a power, and the exponent 1

Power of a power	Repeated multiplication	Single power
$(4^2)^3$	$4^2 \cdot 4^2 \cdot 4^2$	4^6
$(10^4)^4$	$10^4 \cdot 10^4 \cdot 10^4 \cdot 10^4$	10^{16}
$(3^1)^5$	$3 \cdot 3 \cdot 3 \cdot 3 \cdot 3$	3^5

Power of a power: $(a^m)^n = a^{mn}$ **Identity:** $a^1 = a$

Example 1

Two ways to the same value.

$$3^2 \cdot 3^3$$

Which law? Then find the value.

Example 1

Worked both ways.

$$3^2 \cdot 3^3 = 3^{2+3} = 3^5 \quad \text{product of powers: same base, add}$$

$$3^5 = 243 \quad \text{evaluate the single power}$$

$$9 \cdot 27 = 243 \quad \text{check: evaluate each, then multiply}$$

Answer: 243

Your Turn

Same steps, your numbers. Boards up when done.

$$2^3 \cdot 2^4$$

Your Turn

Answer.

$$2^3 \cdot 2^4$$

same base, add: 2^7

$$\text{Answer: } 2^7 = 128$$

Example 2

A negative base and a power of a power.

$$[(-2)^3]^2$$

Which law? Then find the value.

Example 2

Worked.

$$[(-2)^3]^2 = (-2)^{3 \cdot 2} = (-2)^6 \quad \text{power of a power: multiply}$$

$$(-2)^6 = 64 \quad \text{six negative factors — positive}$$

$$(-8)^2 = 64 \quad \text{check: } (-2)^3 = -8, \text{ then square}$$

Answer: 64

Your Turn

Same steps, your numbers. Boards up when done.

$$[(-3)^2]^3$$

Your Turn

Answer.

$$[(-3)^2]^3$$

multiply the exponents: $(-3)^6$

Answer: $(-3)^6 = 729$

Whiteboards • Question 1 of 9

Boards up on three.

$$5^2 \cdot 5^3$$

Answer it.

Write it as one power.

Whiteboards • Question 1 of 9

Answer.

$$5^2 \cdot 5^3$$

same base, so add the exponents

Answer: 5^5

Whiteboards • Question 2 of 9

Boards up on three.

A rectangle is 2^4 centimeters long and 2^2 centimeters wide.

Its perimeter is 40 centimeters.

What is the area of the rectangle, in square centimeters?

Answer it.

Whiteboards • Question 2 of 9

Answer.

A rectangle is 2^4 centimeters long and 2^2 centimeters wide.

Its perimeter is 40 centimeters.

What is the area of the rectangle, in square centimeters?

$$2^4 \cdot 2^2 = 2^6$$

Answer: $2^6 = 64$ square centimeters

Whiteboards • Question 3 of 9

Boards up on three.

$$(2 \cdot 5)^3$$

Answer it.

Find the value.

Whiteboards • Question 3 of 9

Answer.

$$(2 \cdot 5)^3$$

$$2^3 \cdot 5^3 = 8 \cdot 125, \text{ or } 10^3$$

Answer: 1000

Whiteboards • Question 4 of 9

Boards up on three.

$$(3^2)^4$$

Answer it.

Write it as one power.

Whiteboards • Question 4 of 9

Answer.

$$(3^2)^4$$

four copies of 3^2 , so eight 3s

Answer: 3^8

Whiteboards • Question 5 of 9

Boards up on three.

$$4^3 \cdot 4^2$$

Which expression is equivalent?

A. 16^5

B. 8^5

C. 4^6

D. 4^5

Answer it.

Whiteboards · Question 5 of 9

Answer.

$$4^3 \cdot 4^2$$

Which expression is equivalent?

A. 16^5

B. 8^5

C. 4^6

D. 4^5

Answer: D — 4^5

Whiteboards • Question 6 of 9

Boards up on three.

$$(2^3)^2$$

Which expression is equivalent?

A. 2^5

B. 4^6

C. 2^6

D. 2^9

Answer it.

Whiteboards • Question 6 of 9

Answer.

$$(2^3)^2$$

Which expression is equivalent?

A. 2^5

B. 4^6

C. 2^6

D. 2^9

Answer: C — 2^6

Whiteboards • Question 7 of 9

Boards up on three.

$$(-3)^2 \cdot (-3)^1$$

Answer it.

Find the value.

Whiteboards • Question 7 of 9

Answer.

$$(-3)^2 \cdot (-3)^1$$

$(-3)^3$ — three negative factors

Answer: -27

Whiteboards • Question 8 of 9

Boards up on three.

$$\left(\frac{1}{2}\right)^2 \cdot \left(\frac{1}{2}\right)^3$$

Answer it.

Find the value.

Whiteboards • Question 8 of 9

Answer.

$$\left(\frac{1}{2}\right)^2 \cdot \left(\frac{1}{2}\right)^3$$
$$\left(\frac{1}{2}\right)^5$$

Answer: $\frac{1}{32}$

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

Aiyanna says she can use the power of a product law to evaluate $(10 + (-2))^3$. Is she right?

Write your answer in sentences.

Say what the law needs, and show the value.

Whiteboards • Question 9 of 9

Answer.

Aiyanna says she can use the power of a product law to evaluate $(10 + (-2))^3$. Is she right?

the parentheses hold a SUM, not a product

Answer: No — $(10 + (-2))^3 = 8^3 = 512$

Independent Set

Six questions. On your own, in silence.

1. Six questions. Work down the page.
2. Show the step that does the work, not just the answer.
3. Silence until the six minutes are up.

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Evaluate powers — SZQ

Multiplication with exponents — 5SK

Power rule — 8DA

Understanding exponents — BFA

Quotient Laws of Exponents

MA.7.NSO.1.1

I can use the quotient of powers, power of a quotient and zero exponent laws to rewrite and evaluate a numerical expression.

Yesterday: same base, add the exponents.

Today: same base, subtract — and exponent 0 is not 0.

Warm-Up

On your own. Four minutes. No notes.

1. Write as one power: $3^2 \cdot 3^3$
2. $\sqrt[3]{-27}$
3. Solve $2x + 3 < 11$
4. Simplify $\frac{5 \cdot 5 \cdot 5}{5 \cdot 5}$

Warm-Up

Answers.

1. Write as one power: $3^2 \cdot 3^3$ 3^5

2. $\sqrt[3]{-27}$ -3

3. Solve $2x + 3 < 11$ $x < 4$

4. Simplify $\frac{5 \cdot 5 \cdot 5}{5 \cdot 5}$ 5

Notes

Where the rule comes from.

I. Quotient of powers — same base, subtract the exponents

$$\frac{5^6}{5^3} = \frac{5 \cdot 5 \cdot 5 \cdot 5 \cdot 5 \cdot 5}{5 \cdot 5 \cdot 5} = 5 \cdot 5 \cdot 5 = 5^3$$

Quotient of powers: $\frac{a^m}{a^n} = a^{m-n} \quad (a \neq 0)$

Check with values: $\frac{5^6}{5^3} = \frac{15625}{125} = 125 = 5^3$

Notes

II. Power of a quotient — the exponent goes top and bottom

$$\left(\frac{8}{10}\right)^3 = \frac{8}{10} \cdot \frac{8}{10} \cdot \frac{8}{10} = \frac{8 \cdot 8 \cdot 8}{10 \cdot 10 \cdot 10} = \frac{8^3}{10^3}$$

Power of a quotient: $\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m} \quad (b \neq 0)$

Either way: $\frac{8^3}{10^3} = \frac{512}{1000} = 0.512$ and $(0.8)^3 = 0.512$

Notes

III. The zero exponent — two roads to the same answer

2^4	2^3	2^2	2^1	2^0
16	8	4	2	1

$$\frac{2^3}{2^3} = \frac{8}{8} = 1 \quad \text{and} \quad \frac{2^3}{2^3} = 2^{3-3} = 2^0 \quad \text{so} \quad 2^0 = 1$$

Zero exponent: $a^0 = 1$ for every nonzero a

Example 1

A negative base.

$$\frac{(-2)^6}{(-2)^3}$$

Which law? Then find the value.

Example 1

Worked.

$$\frac{(-2)^6}{(-2)^3} = (-2)^{6-3} = (-2)^3 \quad \text{quotient of powers: subtract}$$

$$(-2)^3 = -8 \quad \text{three negative factors}$$

$$64 \div (-8) = -8 \quad \text{check: evaluate each, then divide}$$

Answer: -8

Your Turn

Same steps, your numbers. Boards up when done.

$$\frac{3^6}{3^2}$$

Your Turn

Answer.

$$\frac{3^6}{3^2}$$

same base, subtract: 3^4

Answer: $3^4 = 81$

Example 2

A decimal base inside a quotient.

$$\left(\frac{0.2}{10}\right)^4$$

Find the value.

Example 2

Worked both ways.

$$\left(\frac{0.2}{10}\right)^4 = \frac{0.2^4}{10^4} = \frac{0.0016}{10000} \quad \text{power of a quotient}$$

$$\frac{0.0016}{10000} = 0.00000016 \quad \text{divide by } 10^4: \text{ four places left}$$

$$(0.02)^4 = 0.00000016 \quad \text{check: divide first, then raise}$$

Answer: 0.00000016

Your Turn

Same steps, your numbers. Boards up when done.

$$\left(\frac{3}{4}\right)^3$$

Your Turn

Answer.

$$\left(\frac{3}{4}\right)^3$$

numerator and denominator both cubed

$$\textbf{Answer: } \frac{27}{64}$$

Whiteboards • Question 1 of 9

Boards up on three.

$$\frac{10^5}{10^3}$$

Answer it.

Write it as one power.

Whiteboards • Question 1 of 9

Answer.

$$\frac{10^5}{10^3}$$

same base, subtract the exponents

Answer: 10^2

Whiteboards • Question 2 of 9

Boards up on three.

A website puts 8^4 photos into 8 equal albums.

The site is 30 days old.

How many photos are in each album?

Answer it.

Whiteboards • Question 2 of 9

Answer.

A website puts 8^4 photos into 8 equal albums.

The site is 30 days old.

How many photos are in each album?

$$8^4 \div 8^1 = 8^3$$

Answer: 512

Whiteboards • Question 3 of 9

Boards up on three.

$$\frac{(-6)^6}{(-6)^5}$$

Answer it.

Find the value.

Whiteboards • Question 3 of 9

Answer.

$$\frac{(-6)^6}{(-6)^5}$$
$$(-6)^1$$

Answer: -6

Whiteboards • Question 4 of 9

Boards up on three.

$$\left(\frac{-5}{10}\right)^7$$

Answer it.

Apply the power of a quotient law. Do not simplify.

Whiteboards • Question 4 of 9

Answer.

$$\left(\frac{-5}{10}\right)^7$$

the exponent goes to both

$$\textbf{Answer: } \frac{(-5)^7}{10^7}$$

Whiteboards • Question 5 of 9

Boards up on three.

$$9^0$$

Which is the value?

A. 1

B. 9

C. $\frac{1}{9}$

D. 0

Answer it.

Whiteboards • Question 5 of 9

Answer.

$$9^0$$

Which is the value?

A. 1

B. 9

C. $\frac{1}{9}$

D. 0

Answer: A — 1

Whiteboards • Question 6 of 9

Boards up on three.

$$\frac{4^8}{4^2}$$

Which expression is equivalent?

A. 1^6

B. 4^{10}

C. 4^6

D. 4^4

Answer it.

Whiteboards • Question 6 of 9

Answer.

$$\frac{4^8}{4^2}$$

Which expression is equivalent?

A. 1^6

B. 4^{10}

C. 4^6

D. 4^4

Answer: C — 4^6

Whiteboards • Question 7 of 9

Boards up on three.

$$\frac{0.5^8}{0.5^5}$$

Answer it.

Find the value.

Whiteboards • Question 7 of 9

Answer.

$$\frac{0.5^8}{0.5^5}$$
$$0.5^3$$

Answer: 0.125

Whiteboards • Question 8 of 9

Boards up on three.

$$\left(\frac{10}{11}\right)^0$$

Answer it.

Find the value.

Whiteboards • Question 8 of 9

Answer.

$$\left(\frac{10}{11}\right)^0$$

any nonzero base to the zero power

Answer: 1

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

Hannah cancels: $\frac{2^3}{2^3} = \frac{8}{8} = 1$. Sebastian subtracts: $\frac{2^3}{2^3} = 2^0$.

Who is right?

Write your answer in sentences.

Say what each one did, and what it tells you about 2^0 .

Whiteboards • Question 9 of 9

Answer.

Hannah cancels: $\frac{2^3}{2^3} = \frac{8}{8} = 1$. Sebastian subtracts: $\frac{2^3}{2^3} = 2^0$.

Who is right?

both are right, so 2^0 must equal 1

Answer: Both — $2^0 = 1$

Independent Set

Six questions. On your own, in silence.

1. Six questions. Work down the page.
2. Show the step that does the work, not just the answer.
3. Silence until the six minutes are up.

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Powers with negative bases — WFW

Division with exponents — XNC

Identify equivalent expressions involving whole-number exponents — KBE

Exponential Expressions

MA.7.NSO.1.1

I can apply more than one law of exponents to evaluate an expression and to write an equivalent one.

Yesterday: seven laws, one at a time.

Today: several laws in one expression — the order is yours.

Warm-Up

On your own. Four minutes. No notes.

1. Find the value: $\frac{10^5}{10^3}$
2. Is $\sqrt{50}$ closer to 7 or to 8?
3. Solve $4(x - 2) = 12$
4. Evaluate $2 + 3 \cdot 4^2$

Warm-Up

Answers.

1. Find the value: $\frac{10^5}{10^3}$ **100**
2. Is $\sqrt{50}$ closer to 7 or to 8? **7**
3. Solve $4(x - 2) = 12$ **$x = 5$**
4. Evaluate $2 + 3 \cdot 4^2$ **50**

Notes

Copy the table. It is your reference for the rest of the unit.

I. The seven laws, in one place

Law	Rule	Example
Product of powers	$a^m \cdot a^n = a^{m+n}$	$2^3 \cdot 2^1 = 2^4$
Quotient of powers	$a^m \div a^n = a^{m-n}$	$(-3)^7 \div (-3)^2 = (-3)^5$
Power of a product	$(ab)^m = a^m \cdot b^m$	$(-7 \cdot 3)^2 = (-7)^2 \cdot 3^2$
Power of a quotient	$(a \div b)^m = a^m \div b^m$	$(13 \div 5)^3 = 13^3 \div 5^3$
Power of a power	$(a^m)^n = a^{mn}$	$(6^3)^4 = 6^{12}$
Identity exponent	$a^1 = a$	$15^1 = 15$
Zero exponent	$a^0 = 1$	$127^0 = 1$

Notes

Two students, two orders, one value.

II. More than one law — make a plan, then follow it

$$\text{Winston: } \left(\frac{2^6}{2^4}\right)^3 = (2^2)^3 = 2^6 = 64$$

$$\text{Yvette: } \left(\frac{2^6}{2^4}\right)^3 = \frac{(2^6)^3}{(2^4)^3} = \frac{2^{18}}{2^{12}} = 2^6 = 64$$

Different plans, same value. Any correct sequence of laws gives the same answer.

Notes

III. Order of operations still applies

- A. Parentheses first — including a fraction bar, which is a pair of parentheses in disguise.
- B. Exponents next — the laws of exponents are applied here.
- C. Then multiply and divide, left to right; then add and subtract.
- D. **Never across a + or a -:** $2^3 + 2^2 = 8 + 4 = 12$, **not** 2^5 .

Example 1

A zero exponent inside a product inside a quotient.

$$\frac{4^5 \cdot 4^0}{4^2}$$

Make a plan. Then find the value.

Example 1

Worked — Caroline's plan and Jamir's plan.

$$\frac{4^5 \cdot 4^0}{4^2} = \frac{4^{5+0}}{4^2} = \frac{4^5}{4^2} \quad \text{Caroline: product of powers first}$$

$$\frac{4^5 \cdot 4^0}{4^2} = \frac{4^5 \cdot 1}{4^2} = \frac{4^5}{4^2} \quad \text{Jamir: zero exponent first}$$

$$\frac{4^5}{4^2} = 4^3 = 64 \quad \text{both: quotient of powers, then evaluate}$$

Answer: 64

Your Turn

Same steps, your numbers. Boards up when done.

$$\frac{5^4 \cdot 5^0}{5^2}$$

Your Turn

Answer.

$$\frac{5^4 \cdot 5^0}{5^2}$$

$5^4 \cdot 1$ over 5^2 , so 5^2

Answer: $5^2 = 25$

Example 2

Three laws, a decimal and a negative base.

$$11.6^0 \cdot \frac{(-15)^4}{1.5^4}$$

Make a plan. Then find the value.

Example 2

Worked.

$$11.6^0 \cdot \frac{(-15)^4}{1.5^4} = 1 \cdot \left(\frac{-15}{1.5}\right)^4 \quad \text{zero exponent; power of a quotient, backwards}$$

$$\left(\frac{-15}{1.5}\right)^4 = (-10)^4 \quad \text{divide inside: } -15 \div 1.5 = -10$$

$$(-10)^4 = 10000 \quad \text{four negative factors: positive}$$

Answer: 10,000

Your Turn

Same steps, your numbers. Boards up when done.

$$7^0 \cdot \frac{(-12)^3}{4^3}$$

Your Turn

Answer.

$$7^0 \cdot \frac{(-12)^3}{4^3}$$

$$1 \cdot (-12 \div 4)^3 = (-3)^3$$

Answer: -27

Whiteboards • Question 1 of 9

Boards up on three.

A warehouse holds 6^4 boxes in 6^2 equal rows.

Each box weighs 3 kilograms.

How many boxes are in each row?

Answer it.

Whiteboards • Question 1 of 9

Answer.

A warehouse holds 6^4 boxes in 6^2 equal rows.

Each box weighs 3 kilograms.

How many boxes are in each row?

6^4 over 6^2 , so 6^2

Answer: 36

Whiteboards • Question 2 of 9

Boards up on three.

$$(3^0 \cdot 3^2)^3$$

Answer it.

Find the value.

Whiteboards · Question 2 of 9

Answer.

$$(3^0 \cdot 3^2)^3$$

$$(3^2)^3 = 3^6$$

Answer: 729

Whiteboards • Question 3 of 9

Boards up on three.

$$\frac{(0.5^3)^2}{(0.5)^2}$$

Answer it.

Find the value.

Whiteboards • Question 3 of 9

Answer.

$$\frac{(0.5^3)^2}{(0.5)^2}$$

$$0.5^6 \div 0.5^2 = 0.5^4$$

Answer: 0.0625

Whiteboards • Question 4 of 9

Boards up on three.

$$\left(\frac{2.5^5}{2.5^3}\right)^1$$

Answer it.

Find the value.

Whiteboards • Question 4 of 9

Answer.

$$\left(\frac{2.5^5}{2.5^3} \right)^1$$

2.5^2 — the exponent 1 changes nothing

Answer: 6.25

Whiteboards • Question 5 of 9

Boards up on three.

$$\frac{(2^2 \cdot 3 \cdot 2^0)^6}{2^{10}}$$

Which is the value?

A. $1/4$

B. 2^{22}

C. 0

D. 4

Answer it.

Whiteboards • Question 5 of 9

Answer.

$$\frac{(2^2 \cdot 3 \cdot 2^0)^6}{2^{10}}$$

Which is the value?

A. $1/4$

B. 2^{22}

C. 0

D. 4

Answer: D — 4

Whiteboards • Question 6 of 9

Boards up on three.

$$\frac{0.2^3}{2^3} \cdot 0.1^2$$

Which expression is equivalent?

A. $0.2^0 \cdot 0.1^2$

B. $0.2^3 \cdot 0.1^2$

C. 0.1^5

D. 0.1^6

Answer it.

Whiteboards • Question 6 of 9

Answer.

$$\frac{0.2^3}{2^3} \cdot 0.1^2$$

Which expression is equivalent?

A. $0.2^0 \cdot 0.1^2$

B. $0.2^3 \cdot 0.1^2$

C. 0.1^5

D. 0.1^6

Answer: C — 0.1^5

Whiteboards • Question 7 of 9

Boards up on three.

$$0.1^2 \cdot (0.1^1)^2$$

Answer it.

Find the value.

Whiteboards • Question 7 of 9

Answer.

$$0.1^2 \cdot (0.1^1)^2$$

$$0.1^2 \cdot 0.1^2 = 0.1^4$$

Answer: 0.0001

Whiteboards • Question 8 of 9

Boards up on three.

$$\frac{(-4)^2 \cdot (-4)^1}{(-4)^0}$$

Answer it.

Find the value.

Whiteboards • Question 8 of 9

Answer.

$$\frac{(-4)^2 \cdot (-4)^1}{(-4)^0}$$

$(-4)^3$ over 1

Answer: -64

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

Winston writes $\left(\frac{2^6}{2^4}\right)^3 = (2^2)^3$. Yvette writes $\left(\frac{2^6}{2^4}\right)^3 = \frac{2^{18}}{2^{12}}$.

Do they get the same value? Explain.

Write your answer in sentences.

Name the law each one used first, and give the value.

Whiteboards • Question 9 of 9

Answer.

Winston writes $\left(\frac{2^6}{2^4}\right)^3 = (2^2)^3$. Yvette writes $\left(\frac{2^6}{2^4}\right)^3 = \frac{2^{18}}{2^{12}}$.

Do they get the same value? Explain.

both plans end at 2^6

Answer: Yes — both give $2^6 = 64$

Independent Set

Six questions. On your own, in silence.

1. Six questions. Work down the page.
2. Show the step that does the work, not just the answer.
3. Silence until the six minutes are up.

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Evaluate expressions using properties of exponents — ZE9

Negative Exponent Law

MA.8.NSO.1.3

I can rewrite a power with a negative exponent using a positive exponent, and evaluate it.

Yesterday the exponents stopped at 0.

Today they keep going — and 5^{-2} is not a negative number.

Warm-Up

On your own. Four minutes. No notes.

1. Find the value: $\frac{5^4 \cdot 5^0}{5^2}$
2. Which is greater, $\sqrt{30}$ or 5.5?
3. Solve $-3x = 21$
4. What is the reciprocal of $\frac{3}{4}$? What is the opposite of $\frac{3}{4}$?

Warm-Up

Answers.

1. Find the value: $\frac{5^4 \cdot 5^0}{5^2}$ **25**
2. Which is greater, $\sqrt{30}$ or 5.5? **5.5**
3. Solve $-3x = 21$ **$x = -7$**
4. What is the reciprocal of $\frac{3}{4}$? What is the opposite of $\frac{3}{4}$?
 $\frac{4}{3}$ and $-\frac{3}{4}$

Notes

Copy the table.

I. The pattern keeps going past zero

2^4	2^3	2^2	2^1	2^0	2^{-1}	2^{-2}	2^{-3}
16	8	4	2	1	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$

$$2^{-1} = \frac{1}{2^1}$$

$$2^{-2} = \frac{1}{2^2} = \frac{1}{4}$$

$$2^{-3} = \frac{1}{2^3} = \frac{1}{8}$$

Notes

II. The negative exponent law

$$a^{-n} = \frac{1}{a^n} \quad \frac{1}{a^{-n}} = a^n \quad (a \neq 0)$$

$$5^{-2} = \frac{1}{5^2} = \frac{1}{25} \quad \text{but} \quad -5^2 = -25 \quad \text{and} \quad (-5)^2 = 25$$

A negative exponent means reciprocal. It never means a negative value.

The base keeps its own sign: $(-2)^{-3} = \frac{1}{(-2)^3} = \frac{1}{-8} = -\frac{1}{8}$

Notes

III. The laws still work with negative exponents

$$6^{-2} \cdot 6^5 = \frac{1}{6^2} \cdot 6^5 = \frac{7776}{36} = 216$$

$$\text{or } 6^{-2} \cdot 6^5 = 6^{-2+5} = 6^3 = 216$$

$$\frac{9^{-5}}{9^3} = 9^{-5-3} = 9^{-8} = \frac{1}{9^8}$$

Add, subtract and multiply exponents exactly as before — the exponents are just integers now.

Example 1

Negative exponents top and bottom.

$$\frac{2^{-3}}{9^{-2}}$$

Rewrite with positive exponents, then find the value.

Example 1

Worked.

$$\frac{2^{-3}}{9^{-2}} = \frac{9^2}{2^3} \quad \text{each power swaps sides and loses the minus}$$

$$\frac{9^2}{2^3} = \frac{81}{8} = 10.125 \quad \text{evaluate}$$

Answer: $81/8 = 10.125$

Your Turn

Same steps, your numbers. Boards up when done.

$$\frac{3^{-2}}{5^{-1}}$$

Your Turn

Answer.

$$\frac{3^{-2}}{5^{-1}}$$

swap sides, positive exponents: 5^1 over 3^2

Answer: $\frac{5}{9}$

Example 2

A quotient law crossing zero.

$$\frac{7^0}{7^{-4}}$$

Find the value.

Example 2

Worked both ways.

$$\frac{7^0}{7^{-4}} = 7^{0 - (-4)} = 7^4 \quad \text{quotient of powers: } 0 - (-4) = 4$$

$$\frac{7^0}{7^{-4}} = \frac{1}{7^{-4}} = 7^4 \quad \text{zero exponent, then move } 7^{-4} \text{ up}$$

$$7^4 = 2401 \quad \text{evaluate}$$

Answer: 2,401

Your Turn

Same steps, your numbers. Boards up when done.

$$\frac{(-8)^{-4}}{(-8)^{-6}}$$

Your Turn

Answer.

$$\frac{(-8)^{-4}}{(-8)^{-6}}$$

$$-4 - (-6) = 2, \text{ so } (-8)^2$$

Answer: 64

Whiteboards • Question 1 of 9

Boards up on three.

$$8^{-3}$$

Answer it.

Rewrite with a positive exponent.

Whiteboards • Question 1 of 9

Answer.

$$8^{-3}$$

reciprocal of 8^3

Answer: $\frac{1}{8^3}$

Whiteboards • Question 2 of 9

Boards up on three.

A sheet of paper 30 centimeters long is folded in half 7 times.

Each fold makes the top layer half as big.

What fraction of the whole sheet is the top layer after 7 folds?

Answer it.

Whiteboards • Question 2 of 9

Answer.

A sheet of paper 30 centimeters long is folded in half 7 times.

Each fold makes the top layer half as big.

What fraction of the whole sheet is the top layer after 7 folds?

halved 7 times: $2^{-7} = 1 \text{ over } 2^7$

Answer: $\frac{1}{128}$

Whiteboards • Question 3 of 9

Boards up on three.

$$\frac{1}{10^{-2}}$$

Answer it.

Rewrite with a positive exponent, then find the value.

Whiteboards • Question 3 of 9

Answer.

$$\frac{1}{10^{-2}}$$

the power moves up: 10^2

Answer: $10^2 = 100$

Whiteboards • Question 4 of 9

Boards up on three.

$$(-2)^{-3}$$

Answer it.

Find the value.

Whiteboards • Question 4 of 9

Answer.

$$(-2)^{-3}$$

$$1 \text{ over } (-2)^3 = 1 \text{ over } -8$$

$$\text{Answer: } -\frac{1}{8}$$

Whiteboards • Question 5 of 9

Boards up on three.

$$5^{-2}$$

Which is the value?

A. 1/10

B. 1/25

C. -25

D. -10

Answer it.

Whiteboards • Question 5 of 9

Answer.

$$5^{-2}$$

Which is the value?

A. 1/10

B. 1/25

C. -25

D. -10

Answer: B — 1/25

Whiteboards • Question 6 of 9

Boards up on three.

$$6^{-2} \cdot 6^5$$

Which expression is equivalent?

A. 6^{-10}

B. 36^3

C. 6^{-7}

D. 6^3

Answer it.

Whiteboards · Question 6 of 9

Answer.

$$6^{-2} \cdot 6^5$$

Which expression is equivalent?

A. 6^{-10}

B. 36^3

C. 6^{-7}

D. 6^3

Answer: D — 6^3

Whiteboards • Question 7 of 9

Boards up on three.

$$3^{-4} \cdot 3^4$$

Answer it.

Find the value.

Whiteboards · Question 7 of 9

Answer.

$$3^{-4} \cdot 3^4$$
$$3^0$$

Answer: 1

Whiteboards • Question 8 of 9

Boards up on three.

$$\frac{9^{-5}}{9^3}$$

Answer it.

Write it as one power with a positive exponent.

Whiteboards • Question 8 of 9

Answer.

$$\frac{9^{-5}}{9^3}$$
$$9^{-8}$$

Answer: $\frac{1}{9^8}$

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

Ellen says the value of 5^{-2} is -25 .

Explain why she is wrong, and show the correct value.

Write your answer in sentences.

Use the word reciprocal.

Whiteboards • Question 9 of 9

Answer.

Ellen says the value of 5^{-2} is -25 .

Explain why she is wrong, and show the correct value.

reciprocal, not opposite

Answer: $5^{-2} = 1/5^2 = 1/25$

Independent Set

Six questions. On your own, in silence.

1. Six questions. Work down the page.
2. Show the step that does the work, not just the answer.
3. Silence until the six minutes are up.

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Understanding negative exponents — YBB

Evaluate powers with negative exponents — WGS

Applying Exponent Laws

MA.8.NSO.1.3

I can rewrite a rational base with a negative exponent using a positive exponent, and apply several laws to generate an equivalent expression.

Yesterday: a negative exponent means reciprocal.

Today the base is a fraction — so the reciprocal flips it.

Warm-Up

On your own. Four minutes. No notes.

1. Find the value: 2^{-3}
2. Rewrite with a positive exponent: $\frac{1}{7^{-2}}$
3. Which is larger, 4^3 or 3^4 ?
4. Write the reciprocal of $-\frac{2}{5}$

Warm-Up

Answers.

1. Find the value: 2^{-3} $\frac{1}{8}$
2. Rewrite with a positive exponent: $\frac{1}{7^{-2}}$ 7^2
3. Which is larger, 4^3 or 3^4 ? 3^4 (**81 > 64**)
4. Write the reciprocal of $-\frac{2}{5}$ $-\frac{5}{2}$

Notes

Copy the table.

I. The pattern with a fraction base

Expression	Expanded	Value
$(2/5)^2$	$2/5 \cdot 2/5$	$4/25$
$(2/5)^1$	$2/5$	$2/5$
$(2/5)^0$	1	1
$(2/5)^{-1}$	$5/2$	$5/2$
$(2/5)^{-2}$	$5/2 \cdot 5/2$	$25/4$

Notes

II. Flip the base, make the exponent positive

$$\left(\frac{a}{b}\right)^{-n} = \left(\frac{b}{a}\right)^n \quad (a, b \neq 0)$$

$$\left(\frac{2}{5}\right)^{-2} = \left(\frac{5}{2}\right)^2 = \frac{25}{4} \quad \left(-\frac{2}{3}\right)^{-5} = \left(-\frac{3}{2}\right)^5$$

The sign stays with the fraction. Flip the numbers, not the sign.

A whole number is a fraction over 1: $4^{-3} = \left(\frac{4}{1}\right)^{-3} = \left(\frac{1}{4}\right)^3$

Notes

III. Several laws at once — the plan is still yours

$$\left(\frac{2.5^5}{2.5^3}\right)^{-1} = (2.5^2)^{-1} = 2.5^{-2} = \left(\frac{5}{2}\right)^{-2} = \left(\frac{2}{5}\right)^2 = \frac{4}{25}$$

Any correct order of laws gives the same value. Finish with a positive exponent.

Example 1

Flip, then evaluate.

$$\left(-\frac{7}{8}\right)^{-2}$$

Rewrite with a positive exponent, then find the value.

Example 1

Worked.

$$\left(-\frac{7}{8}\right)^{-2} = \left(-\frac{8}{7}\right)^2 \quad \text{flip the base; the sign stays}$$

$$\left(-\frac{8}{7}\right)^2 = \frac{64}{49} \quad \text{even power: positive}$$

Answer: 64/49

Your Turn

Same steps, your numbers. Boards up when done.

$$\left(-\frac{5}{2}\right)^{-3}$$

Your Turn

Answer.

$$\left(-\frac{5}{2}\right)^{-3}$$

$(-2/5)^3$ — odd power, negative

Answer: $-\frac{8}{125}$

Example 2

Two laws, then a flip.

$$\left(\frac{5}{8}\right)^{-2} \cdot \left(\frac{5}{8}\right)^4$$

Write an equivalent expression with one base and a positive exponent.

Example 2

Worked.

$$\left(\frac{5}{8}\right)^{-2} \cdot \left(\frac{5}{8}\right)^4 = \left(\frac{5}{8}\right)^{-2+4} = \left(\frac{5}{8}\right)^2 \quad \text{product of powers}$$

$$\left(\frac{5}{8}\right)^2 = \frac{25}{64} \quad \text{already positive: no flip}$$

Answer: $(5/8)^2 = 25/64$

Your Turn

Same steps, your numbers. Boards up when done.

$$\left(-\frac{5}{4}\right)^{-7} \cdot \left(-\frac{5}{4}\right)^{-3}$$

Your Turn

Answer.

$$\left(-\frac{5}{4}\right)^{-7} \cdot \left(-\frac{5}{4}\right)^{-3}$$

exponent -10 , then flip: $(-4/5)^{10}$

Answer: $\left(-\frac{4}{5}\right)^{10}$

Whiteboards • Question 1 of 9

Boards up on three.

A copier is set to shrink a page to $\frac{2}{3}$ of its size.

A page 6 inches wide is copied 4 times in a row, each copy from the last.

What fraction of the original page's size is the last copy?

Answer it.

Whiteboards • Question 1 of 9

Answer.

A copier is set to shrink a page to $\frac{2}{3}$ of its size.

A page 6 inches wide is copied 4 times in a row, each copy from the last.

What fraction of the original page's size is the last copy?

$$\left(\frac{2}{3}\right)^4 = \frac{2^4}{3^4}$$

Answer: $\frac{16}{81}$

Whiteboards • Question 2 of 9

Boards up on three.

$$\left(\frac{3}{-5}\right)^{-9}$$

Answer it.

Rewrite with a positive exponent.

Whiteboards • Question 2 of 9

Answer.

$$\left(\frac{3}{-5}\right)^{-9}$$

flip the numbers; the sign stays in the fraction

Answer: $\left(\frac{-5}{3}\right)^9$

Whiteboards • Question 3 of 9

Boards up on three.

$$\left(-\frac{8}{9}\right)^{-2}$$

Answer it.

Find the value.

Whiteboards • Question 3 of 9

Answer.

$$\left(-\frac{8}{9}\right)^{-2}$$

$$(-9/8)^2 = 81/64$$

$$\text{Answer: } \frac{81}{64}$$

Whiteboards • Question 4 of 9

Boards up on three.

$$\left(\frac{21}{4}\right)^{-5}$$

Answer it.

Rewrite with a positive exponent.

Whiteboards • Question 4 of 9

Answer.

$$\left(\frac{21}{4}\right)^{-5}$$

flip

Answer: $\left(\frac{4}{21}\right)^5$

Whiteboards • Question 5 of 9

Boards up on three.

$$\left(-\frac{7}{8}\right)^{-3}$$

Which expression is equivalent?

A. $(7/8)^3$

B. $(-8/7)^3$

C. $(-7/8)^3$

D. $(8/7)^3$

Answer it.

Whiteboards • Question 5 of 9

Answer.

$$\left(-\frac{7}{8}\right)^{-3}$$

Which expression is equivalent?

A. $(7/8)^3$

B. $(-8/7)^3$

C. $(-7/8)^3$

D. $(8/7)^3$

Answer: B — $(-8/7)^3$

Whiteboards • Question 6 of 9

Boards up on three.

$$\left(\frac{5}{8}\right)^{-2} \div \left(\frac{5}{8}\right)^{-4}$$

Which expression is equivalent?

A. $(8/5)^2$

B. $(5/8)^{-6}$

C. $(5/8)^2$

D. $(5/8)^8$

Answer it.

Whiteboards • Question 6 of 9

Answer.

$$\left(\frac{5}{8}\right)^{-2} \div \left(\frac{5}{8}\right)^{-4}$$

Which expression is equivalent?

A. $(8/5)^2$

B. $(5/8)^{-6}$

C. $(5/8)^2$

D. $(5/8)^8$

Answer: C — $(5/8)^2$

Whiteboards • Question 7 of 9

Boards up on three.

$$\left(\frac{2}{5}\right)^{-1} \cdot \left(\frac{5}{2}\right)^1$$

Answer it.

Find the value.

Whiteboards • Question 7 of 9

Answer.

$$\left(\frac{2}{5}\right)^{-1} \cdot \left(\frac{5}{2}\right)^1$$

$$(5/2)^1 \cdot (5/2)^1 = (5/2)^2$$

$$\text{Answer: } \frac{25}{4}$$

Whiteboards • Question 8 of 9

Boards up on three.

$$\left[\left(\frac{2}{3}\right)^5 \cdot \left(\frac{2}{3}\right)^{-2}\right]^3$$

Answer it.

Write it with one base and one positive exponent.

Whiteboards • Question 8 of 9

Answer.

$$\left[\left(\frac{2}{3} \right)^5 \cdot \left(\frac{2}{3} \right)^{-2} \right]^3$$

$(2/3)^3$ inside, then power of a power

Answer: $\left(\frac{2}{3} \right)^9$

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

Marco says $\left(-\frac{2}{3}\right)^{-5}$ can be rewritten as $\left(\frac{2}{3}\right)^5$

because 'two negatives make a positive.' Is he right?

Write your answer in sentences.

Say what the negative exponent does, and what it does not do. Give the correct rewrite.

Whiteboards • Question 9 of 9

Answer.

Marco says $\left(-\frac{2}{3}\right)^{-5}$ can be rewritten as $\left(\frac{2}{3}\right)^5$

because 'two negatives make a positive.' Is he right?

the exponent flips the fraction; it never touches the sign

Answer: No — $(-2/3)^{-5} = (-3/2)^5$

Independent Set

Six questions. On your own, in silence.

1. Six questions. Work down the page.
2. Show the step that does the work, not just the answer.
3. Silence until the six minutes are up.

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Identify equivalent expressions involving exponents I — VLM

Evaluating and Equivalent Expressions

MA.8.NSO.1.3

I can evaluate an expression with integer exponents using the laws, and decide whether two expressions with different bases are equivalent.

So far: every law, every kind of base, every kind of exponent.

Today: use all of it at once — and rewrite bases to compare.

Warm-Up

On your own. Four minutes. No notes.

1. Rewrite with a positive exponent, then evaluate: $\left(\frac{2}{3}\right)^{-2}$
2. Find the value: $\frac{1}{5^{-2}}$
3. Find the value: $\frac{10^6}{10^4}$
4. Write 64 as a power of 2, of 4, and of 8.

Warm-Up

Answers.

1. Rewrite with a positive exponent, then evaluate: $\left(\frac{2}{3}\right)^{-2}$
 $\left(\frac{3}{2}\right)^2 = \frac{9}{4}$
2. Find the value: $\frac{1}{5^{-2}}$ **25**
3. Find the value: $\frac{10^6}{10^4}$ **100**
4. Write 64 as a power of 2, of 4, and of 8. **$2^6, 4^3, 8^2$**

Notes

I. Two students, two plans, one value

$$\text{Melanie: } \frac{5^{-2} \cdot 5^5}{5} = \frac{5^3}{5} = 5^2 = 25$$

$$\text{Adonis: } \frac{5^{-2} \cdot 5^5}{5} = \frac{\frac{1}{25} \cdot 3125}{5} = \frac{125}{5} = 25$$

Apply the laws first, evaluate last. The value is the same; the arithmetic is smaller.

Notes

Copy the worked line.

II. Every law, one expression

$$\frac{(6^2)^{-3}}{6^{-5}} = \frac{6^{-6}}{6^{-5}} = 6^{-6 - (-5)} = 6^{-1} = \frac{1}{6}$$

Find the step that broke:

$$\left(\frac{1}{5}\right)^4 \cdot \left(\frac{1}{5}\right)^{-2} = \left(\frac{1}{5}\right)^{-8} \quad \text{— wrong: multiplied; should be } \left(\frac{1}{5}\right)^2$$

$$\left(\frac{3}{2}\right)^{-3} = \frac{2^{-3}}{3^{-3}} \quad \text{— wrong: flipped, kept the minus; should be } \left(\frac{2}{3}\right)^3$$

Notes

III. Different bases, same value — rewrite to a common base

Base	Square	Cube	Fourth
2	$2^2 = 4$	$2^3 = 8$	$2^4 = 16$
3	$3^2 = 9$	$3^3 = 27$	$3^4 = 81$
5	$5^2 = 25$	$5^3 = 125$	$5^4 = 625$

$$8^5 = (2^3)^5 = 2^{15} \qquad 9^4 = (3^2)^4 = 3^8$$

$$2^3 \cdot 4^3 = 2^3 \cdot (2^2)^3 = 2^3 \cdot 2^6 = 2^9$$

Notes

IV. Testing equivalence: Xavier's claim

$$\left(\frac{1}{4}\right)^2 \cdot 2^{-3} = 2^{-4} \cdot 2^{-3} = 2^{-7} \qquad \left(\frac{1}{2}\right)^4 \cdot 4^{-1} = 2^{-4} \cdot 2^{-2} = 2^{-6}$$

Same base, then compare exponents.

$2^{-7} \neq 2^{-6}$ — so Xavier is wrong.

Example 1

Evaluate.

$$\left(\frac{3}{4}\right)^{-2} \cdot 3$$

Find the value.

Example 1

Worked.

$$\left(\frac{3}{4}\right)^{-2} \cdot 3 = \left(\frac{4}{3}\right)^2 \cdot 3 \quad \text{negative exponent: flip}$$

$$\frac{16}{9} \cdot 3 = \frac{16}{3} \quad \text{power of a quotient, then multiply}$$

Answer: 16/3

Your Turn

Same steps, your numbers. Boards up when done.

$$\left(\frac{2}{5}\right)^{-2} \cdot 2$$

Your Turn

Answer.

$$\left(\frac{2}{5}\right)^{-2} \cdot 2$$

$$(5/2)^2 \cdot 2 = 25/4 \cdot 2$$

$$\textbf{Answer: } \frac{25}{2}$$

Example 2

Equivalent or not?

Are $3^2 \cdot 6^3$ and $\frac{(3 \cdot 2)^5}{2^2}$ equivalent?

Rewrite both with the bases 3 and 2, then compare.

Example 2

Worked.

$$3^2 \cdot 6^3 = 3^2 \cdot 3^3 \cdot 2^3 = 3^5 \cdot 2^3 \quad \text{power of a product, product of powers}$$

$$\frac{(3 \cdot 2)^5}{2^2} = \frac{3^5 \cdot 2^5}{2^2} = 3^5 \cdot 2^3 \quad \text{power of a product, quotient of powers}$$

$$3^5 \cdot 2^3 = 3^5 \cdot 2^3 \quad \text{same bases, same exponents}$$

Answer: Yes — both are $3^5 \cdot 2^3 = 1944$

Your Turn

Same steps, your numbers. Boards up when done.

Are $2^2 \cdot 10^3$ and $\frac{(2 \cdot 5)^5}{5^2}$ equivalent?

Your Turn

Answer.

Are $2^2 \cdot 10^3$ and $\frac{(2 \cdot 5)^5}{5^2}$ equivalent?

both are $2^5 \cdot 5^3$

Answer: Yes — $2^5 \cdot 5^3 = 4000$

Whiteboards • Question 1 of 9

Boards up on three.

$$\frac{(-7)^5}{(-7)^0 \cdot (-7)^2}$$

Answer it.

Find the value.

Whiteboards • Question 1 of 9

Answer.

$$\frac{(-7)^5}{(-7)^0 \cdot (-7)^2}$$

$(-7)^5$ over $(-7)^2$, so $(-7)^3$

Answer: -343

Whiteboards • Question 2 of 9

Boards up on three.

$$\frac{\left(\frac{3}{4}\right)^{-3} \cdot \left(\frac{3}{4}\right)^0}{\left(\frac{3}{4}\right)^{-2}}$$

Answer it.

Find the value.

Whiteboards • Question 2 of 9

Answer.

$$\frac{\left(\frac{3}{4}\right)^{-3} \cdot \left(\frac{3}{4}\right)^0}{\left(\frac{3}{4}\right)^{-2}}$$

$$-3 - (-2) = -1, \text{ so } (3/4)^{-1}$$

$$\text{Answer: } \frac{4}{3}$$

Whiteboards • Question 3 of 9

Boards up on three.

$$\frac{(2^{-1})^3 \cdot \left(\frac{1}{2}\right)^{-4}}{2 \cdot 2^2}$$

Answer it.

Find the value.

Whiteboards • Question 3 of 9

Answer.

$$\frac{(2^{-1})^3 \cdot \left(\frac{1}{2}\right)^{-4}}{2 \cdot 2^2}$$

$$2^{-3} \cdot 2^4 \text{ over } 2^3 = 2^{-2}$$

$$\text{Answer: } \frac{1}{4}$$

Whiteboards • Question 4 of 9

Boards up on three.

One flash drive holds 2^{15} megabytes.

Another holds 8^5 megabytes. The first one costs 45 dollars.

Which holds more — or is it the same? Show why.

Answer it.

Whiteboards • Question 4 of 9

Answer.

One flash drive holds 2^{15} megabytes.

Another holds 8^5 megabytes. The first one costs 45 dollars.

Which holds more — or is it the same? Show why.

$$8 = 2^3, \text{ so } 8^5 = (2^3)^5$$

Answer: The same: $8^5 = (2^3)^5 = 2^{15}$

Whiteboards • Question 5 of 9

Boards up on three.

$$9^4$$

Which expression is equivalent?

A. 3^{16}

B. 3^4

C. 3^8

D. 3^6

Answer it.

Whiteboards • Question 5 of 9

Answer.

$$9^4$$

Which expression is equivalent?

A. 3^{16}

B. 3^4

C. 3^8

D. 3^6

Answer: C — 3^8

Whiteboards • Question 6 of 9

Boards up on three.

$$\left(\frac{1}{4}\right)^2 \cdot 2^{-3}$$

Which expression is equivalent?

A. 2^{-7}

B. 2^7

C. 2^{-5}

D. 2^{-12}

Answer it.

Whiteboards • Question 6 of 9

Answer.

$$\left(\frac{1}{4}\right)^2 \cdot 2^{-3}$$

Which expression is equivalent?

A. 2^{-7}

B. 2^7

C. 2^{-5}

D. 2^{-12}

Answer: A — 2^{-7}

Whiteboards • Question 7 of 9

Boards up on three.

$$2^3 \cdot 4^3$$

Answer it.

Write it as one power of 2.

Whiteboards • Question 7 of 9

Answer.

$$2^3 \cdot 4^3$$

$$4^3 = (2^2)^3 = 2^6$$

Answer: 2^9

Whiteboards • Question 8 of 9

Boards up on three.

$$4^7 \cdot 4^{-2}$$

Answer it.

Write it as one power of 2.

Whiteboards • Question 8 of 9

Answer.

$$4^7 \cdot 4^{-2}$$

$$4^5 = (2^2)^5$$

Answer: 2^{10}

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

Xavier says $\left(\frac{1}{4}\right)^2 \cdot 2^{-3}$ and $\left(\frac{1}{2}\right)^4 \cdot 4^{-1}$ are equivalent

because both can be written as 2^{-7} . Do you agree?

Write your answer in sentences.

Rewrite each with base 2, and say what you find.

Whiteboards • Question 9 of 9

Answer.

Xavier says $\left(\frac{1}{4}\right)^2 \cdot 2^{-3}$ and $\left(\frac{1}{2}\right)^4 \cdot 4^{-1}$ are equivalent

because both can be written as 2^{-7} . Do you agree?

2^{-7} and 2^{-6}

Answer: No — the second is 2^{-6} , not 2^{-7}

Independent Set

Six questions. On your own, in silence.

1. Six questions. Work down the page.
2. Show the step that does the work, not just the answer.
3. Silence until the six minutes are up.

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Evaluate expressions using properties of exponents — 6T7

Identify equivalent expressions involving exponents II — QDM

Exponent Laws with Variable Bases

MA.8.AR.1.1

I can multiply and divide monomials, and raise a monomial to a power, by applying the laws of exponents to variable bases.

Yesterday the bases were all numbers.

Today the base is a letter. Every law still works.

Warm-Up

On your own. Four minutes. No notes.

1. Find the value: $(2^3)^2 \cdot 2^{-4}$
2. Write as one power: $\frac{6^7}{6^5}$
3. $\sqrt{50}$ is between which two whole numbers?
4. Write with an exponent: $y \cdot y \cdot y \cdot y \cdot y$

Warm-Up

Answers.

1. Find the value: $(2^3)^2 \cdot 2^{-4}$ **4**
2. Write as one power: $\frac{6^7}{6^5}$ **$6^2 = 36$**
3. $\sqrt{50}$ is between which two whole numbers? **7 and 8**
4. Write with an exponent: $y \cdot y \cdot y \cdot y \cdot y$ **y^5**

Notes

Copy both lines.

I. Product of powers — the base is a letter now

$$x^2 \cdot x^3 = (x \cdot x)(x \cdot x \cdot x) = x^5 \qquad x^m \cdot x^n = x^{m+n}$$

$$3x^2 \cdot 5x^4 = (3 \cdot 5)(x^2 \cdot x^4) = 15x^6$$

Multiply the coefficients. Add the exponents of like bases.

Different variables stay separate: $x^2y \cdot x^3y^4 = x^5y^5$

Notes

II. Power of a power, power of a product

$$(x^3)^4 = x^3 \cdot x^3 \cdot x^3 \cdot x^3 = x^{12}$$

$$(x^m)^n = x^{mn}$$

$$(2y^2)^3 = 2^3 \cdot (y^2)^3 = 8y^6$$

$$(15xy^2)^3 = 3,375x^3y^6$$

Power of a power: multiply the exponents. Power of a product: raise every factor.

Notes

III. Quotient of powers — dividing monomials

$$\frac{x^7}{x^2} = \frac{x \cdot x \cdot x \cdot x \cdot x \cdot x \cdot x}{x \cdot x} = x^5$$

$$\frac{x^m}{x^n} = x^{m-n}$$

$$\frac{12a^5}{4a^2} = 3a^3$$

$$\frac{m^6n^4}{m^2n^4} = m^4n^0 = m^4$$

Divide the coefficients. Subtract the exponents of like bases.

Example 1

Multiplying two monomials.

$$(4x^3y)(-2x^2y^5)$$

Write an equivalent expression as a single monomial.

Example 1

Worked.

$$(4x^3y)(-2x^2y^5) = (4)(-2) \cdot x^3x^2 \cdot y^1y^5 \quad \text{regroup: coefficients, } x\text{'s, } y\text{'s}$$
$$= -8 \cdot x^5 \cdot y^6 = -8x^5y^6 \quad \text{multiply coefficients; add exponents}$$

Answer: $-8x^5y^6$

Your Turn

Same steps, your numbers. Boards up when done.

$$(5a^2b^4)(3ab^2)$$

Your Turn

Answer.

$$(5a^2b^4)(3ab^2)$$

$$5 \cdot 3, a^2 \cdot a^1, b^4 \cdot b^2$$

$$\textbf{Answer: } 15a^3b^6$$

Example 2

A power of a product, then a quotient.

$$\frac{(3m^2n)^3}{9m^4n}$$

Write an equivalent expression as a single monomial.

Example 2

Worked.

$$(3m^2n)^3 = 27m^6n^3 \quad \text{power of a product: raise all three factors}$$

$$\frac{27m^6n^3}{9m^4n} = 3 \cdot m^{6-4} \cdot n^{3-1} = 3m^2n^2 \quad \text{divide coefficients; subtract exponents}$$

Answer: $3m^2n^2$

Your Turn

Same steps, your numbers. Boards up when done.

$$\frac{(2p^3)^4}{8p^5}$$

Your Turn

Answer.

$$\frac{(2p^3)^4}{8p^5}$$

$16p^{12}$ over $8p^5$

Answer: $2p^7$

Whiteboards • Question 1 of 9

Boards up on three.

$$x^5 \cdot x^8$$

Answer it.

Write as a single power.

Whiteboards • Question 1 of 9

Answer.

$$x^5 \cdot x^8$$

like bases: add the exponents

$$\textbf{Answer: } x^{13}$$

Whiteboards • Question 2 of 9

Boards up on three.

$$(y^4)^3$$

Answer it.

Write as a single power.

Whiteboards • Question 2 of 9

Answer.

$$(y^4)^3$$

power of a power: multiply

Answer: y^{12}

Whiteboards • Question 3 of 9

Boards up on three.

A cube has edges $2x^3$ centimeters long.

Its surface area is $24x^6$ square centimeters.

What is the volume of the cube, in cubic centimeters?

Answer it.

Whiteboards • Question 3 of 9

Answer.

A cube has edges $2x^3$ centimeters long.

Its surface area is $24x^6$ square centimeters.

What is the volume of the cube, in cubic centimeters?

$(2x^3)^3$: raise the 2 as well

Answer: $8x^9$

Whiteboards • Question 4 of 9

Boards up on three.

$$(3a^2b)(4ab^5)$$

Answer it.

Write as a single monomial.

Whiteboards • Question 4 of 9

Answer.

$$(3a^2b)(4ab^5)$$

$$3 \cdot 4, a^2 \cdot a, b \cdot b^5$$

$$\textbf{Answer: } 12a^3b^6$$

Whiteboards • Question 5 of 9

Boards up on three.

$$(5m^3)^2$$

Which expression is equivalent?

A. $25m^6$

B. $10m^6$

C. $25m^5$

D. $5m^6$

Answer it.

Whiteboards • Question 5 of 9

Answer.

$$(5m^3)^2$$

Which expression is equivalent?

A. $25m^6$

B. $10m^6$

C. $25m^5$

D. $5m^6$

Answer: A — $25m^6$

Whiteboards • Question 6 of 9

Boards up on three.

$$\frac{12x^8}{3x^2}$$

Which expression is equivalent?

A. $4x^{10}$

B. $4x^6$

C. $9x^6$

D. $4x^4$

Answer it.

Whiteboards • Question 6 of 9

Answer.

$$\frac{12x^8}{3x^2}$$

Which expression is equivalent?

A. $4x^{10}$

B. $4x^6$

C. $9x^6$

D. $4x^4$

Answer: B — $4x^6$

Whiteboards • Question 7 of 9

Boards up on three.

$$\frac{m^6 n^4}{m^2 n^4}$$

Answer it.

Write as a single monomial.

Whiteboards • Question 7 of 9

Answer.

$$\frac{m^6 n^4}{m^2 n^4}$$

$$m^4 \cdot n^0, \text{ and } n^0 = 1$$

Answer: m^4

Whiteboards • Question 8 of 9

Boards up on three.

Find the value of n :

$$x^3 \cdot x^n = x^{11}$$

Answer it.

Which law applies?

Whiteboards • Question 8 of 9

Answer.

Find the value of n :

$$x^3 \cdot x^n = x^{11}$$

$$3 + n = 11$$

Answer: $n = 8$

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

Rachel writes $(15xy^2)^3$ as $15 \cdot 15 \cdot 15 \cdot x \cdot x \cdot x \cdot y \cdot y \cdot y \cdot y \cdot y \cdot y$.

Justina writes it as $3,375x^3y^6$. Who is correct? Explain.

Write your answer in sentences.

Say what each of them did.

Whiteboards • Question 9 of 9

Answer.

Rachel writes $(15xy^2)^3$ as $15 \cdot 15 \cdot 15 \cdot x \cdot x \cdot x \cdot y \cdot y \cdot y \cdot y \cdot y \cdot y$.

Justina writes it as $3,375x^3y^6$. Who is correct? Explain.

both — Justina's is Rachel's product, multiplied out

Answer: Both are correct; $15^3 = 3,375$, $x \cdot x \cdot x = x^3$, and six y 's make y^6 .

Independent Set

Six questions. On your own, in silence.

1. Six questions. Work down the page.
2. Show the step that does the work, not just the answer.
3. Silence until the six minutes are up.

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Multiply powers: variable bases — TR9

Powers of a power: variable bases — 2CU

Divide monomials with positive exponents — C8L

Negative Exponents with Variable Bases

MA.8.AR.1.1

I can rewrite an algebraic expression with negative exponents using only positive exponents, and apply several laws of exponents to a monomial expression.

Yesterday every exponent on a letter was positive.

Today x^3 arrives — and it is not a negative number.

Warm-Up

On your own. Four minutes. No notes.

1. Write as a single monomial: $(3x^2)^3$
2. Find the value: 5^{-2}
3. Which is greater, $\sqrt{80}$ or 9?
4. What is the reciprocal of $\frac{2}{3}$?

Warm-Up

Answers.

1. Write as a single monomial: $(3x^2)^3$ **$27x^6$**
2. Find the value: 5^{-2} **$1/25$**
3. Which is greater, $\sqrt{80}$ or 9? **9**
4. What is the reciprocal of $\frac{2}{3}$? **$\frac{3}{2}$**

Notes

Copy the table.

I. The pattern, with a letter

$\mathbf{x^4}$	$\mathbf{x^3}$	$\mathbf{x^2}$	$\mathbf{x^1}$	$\mathbf{x^0}$	$\mathbf{x^{-1}}$	$\mathbf{x^{-2}}$	$\mathbf{x^{-3}}$
$\mathbf{x \cdot x \cdot x \cdot x}$	$\mathbf{x \cdot x \cdot x}$	$\mathbf{x \cdot x}$	$\mathbf{x}$	$\mathbf{1}$	$\mathbf{1/x}$	$\mathbf{1/x^2}$	$\mathbf{1/x^3}$

$$x^{-n} = \frac{1}{x^n} \qquad \frac{1}{x^{-n}} = x^n \qquad (x \neq 0)$$

Notes

II. Moving factors across the fraction bar

$$3x^{-2} = \frac{3}{x^2}$$

$$y^{-3}z^{-4} = \frac{1}{y^3z^4}$$

$$\frac{a^2}{b^{-3}} = a^2b^3$$

$$\frac{5x^{-4}y^2}{z^{-1}} = \frac{5y^2z}{x^4}$$

Only the factor with the negative exponent moves. Its exponent turns positive.

Notes

III. The laws with negative exponents

$$x^4 \cdot x^{-6} = x^{4+(-6)} = x^{-2} = \frac{1}{x^2} \qquad \frac{m^3}{m^7} = m^{3-7} = m^{-4} = \frac{1}{m^4}$$

$$(2y^{-3})^2 = 2^2 \cdot y^{-6} = \frac{4}{y^6}$$

Apply the law. Then rewrite so that every exponent is positive.

Example 1

Multiplying monomials with negative exponents.

$$(2x^{-3}y^2)(6x^5y^{-4})$$

Write an equivalent expression with positive exponents only.

Example 1

Worked.

$$(2x^{-3}y^2)(6x^5y^{-4}) = 12 \cdot x^{-3+5} \cdot y^{2+(-4)} \quad \begin{array}{l} \text{multiply coefficients;} \\ \text{add exponents} \end{array}$$

$$= 12x^2y^{-2} = \frac{12x^2}{y^2} \quad \text{move } y^{-2} \text{ down; the 12 and the } x^2 \text{ stay up}$$

Answer: $12x^2/y^2$

Your Turn

Same steps, your numbers. Boards up when done.

$$(4a^{-2}b)(a^5b^{-3})$$

Your Turn

Answer.

$$(4a^{-2}b)(a^5b^{-3})$$

$$4 \cdot a^3 \cdot b^{-2}$$

$$\textbf{Answer: } \frac{4a^3}{b^2}$$

Example 2

Several laws in one expression — the state guide's item.

$$\left(\frac{a^3}{4b^{-7}}\right)^5$$

Write an equivalent expression with positive exponents only.

Example 2

Worked.

$$\left(\frac{a^3}{4b^{-7}}\right)^5 = \frac{a^{15}}{4^5 b^{-35}} \quad \text{raise every factor: } a^3, 4, b^{-7}$$
$$= \frac{a^{15} b^{35}}{1,024} \quad 4^5 = 1,024; b^{-35} \text{ moves up as } b^{35}$$

Answer: $a^{15}b^{35}/1,024$

Your Turn

Same steps, your numbers. Boards up when done.

$$\left(\frac{x^2}{3y^{-2}}\right)^3$$

Your Turn

Answer.

$$\left(\frac{x^2}{3y^{-2}} \right)^3$$

x^6 over $27y^6$, then move the y up

Answer: $\frac{x^6 y^6}{27}$

Whiteboards • Question 1 of 9

Boards up on three.

$$x^{-4}$$

Answer it.

Rewrite with a positive exponent.

Whiteboards • Question 1 of 9

Answer.

$$x^{-4}$$

reciprocal of x^4

Answer: $\frac{1}{x^4}$

Whiteboards • Question 2 of 9

Boards up on three.

$$\frac{1}{y^{-5}}$$

Answer it.

Rewrite with a positive exponent.

Whiteboards • Question 2 of 9

Answer.

$$\frac{1}{y^{-5}}$$

the power moves up

Answer: y^5

Whiteboards • Question 3 of 9

Boards up on three.

The brightness of a lamp from x meters away is $3x^{-2}$ units.

The lamp is 15 centimeters tall.

How bright is the lamp from 2 meters away?

Answer it.

Whiteboards • Question 3 of 9

Answer.

The brightness of a lamp from x meters away is $3x^{-2}$ units.

The lamp is 15 centimeters tall.

How bright is the lamp from 2 meters away?

$$3 \cdot 2^{-2} = 3 \text{ over } 2^2$$

Answer: $\frac{3}{4}$

Whiteboards • Question 4 of 9

Boards up on three.

$$y^{-3}z^{-4}$$

Answer it.

Rewrite with positive exponents.

Whiteboards • Question 4 of 9

Answer.

$$y^{-3}z^{-4}$$

both factors move down

$$\text{Answer: } \frac{1}{y^3 z^4}$$

Whiteboards • Question 5 of 9

Boards up on three.

$$x^4 \cdot x^{-6}$$

Which expression is equivalent?

A. $-x^2$

B. $1/x^{24}$

C. $1/x^2$

D. x^{10}

Answer it.

Whiteboards • Question 5 of 9

Answer.

$$x^4 \cdot x^{-6}$$

Which expression is equivalent?

A. $-x^2$

B. $1/x^{24}$

C. $1/x^2$

D. x^{10}

Answer: C — $1/x^2$

Whiteboards • Question 6 of 9

Boards up on three.

$$(2x^{-3})^2$$

Which expression is equivalent?

A. $4/x$

B. $1/(4x^6)$

C. $2/x^6$

D. $4/x^6$

Answer it.

Whiteboards • Question 6 of 9

Answer.

$$(2x^{-3})^2$$

Which expression is equivalent?

A. $4/x$

B. $1/(4x^6)$

C. $2/x^6$

D. $4/x^6$

Answer: D — $4/x^6$

Whiteboards • Question 7 of 9

Boards up on three.

$$\frac{m^3}{m^7}$$

Answer it.

Write with a positive exponent.

Whiteboards • Question 7 of 9

Answer.

$$\frac{m^3}{m^7}$$

m^{-4} — then rewrite

Answer: $\frac{1}{m^4}$

Whiteboards • Question 8 of 9

Boards up on three.

Find the value of n :

$$x^{-5} \cdot x^n = x^2$$

Answer it.

Which law applies?

Whiteboards • Question 8 of 9

Answer.

Find the value of n :

$$x^{-5} \cdot x^n = x^2$$

$$-5 + n = 2$$

Answer: $n = 7$

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

How are x^3 and x^{-3} related?

Explain, and give the value of each when $x = 2$.

Write your answer in sentences.

Use the word reciprocal.

Whiteboards · Question 9 of 9

Answer.

How are x^3 and x^{-3} related?

Explain, and give the value of each when $x = 2$.

reciprocals: 8 and $1/8$

Answer: x^{-3} is the reciprocal of x^3 . When $x = 2$: $x^3 = 8$ and $x^{-3} = 1/8$.

Independent Set

Six questions. On your own, in silence.

1. Six questions. Work down the page.
2. Show the step that does the work, not just the answer.
3. Silence until the six minutes are up.

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Variables with negative exponents — 2NL

Multiply and divide powers: variable bases — 85P

Writing Large Numbers in Scientific Notation

MA.8.NSO.1.4

I can write a large number in scientific notation, read one back into standard form, and find how many times larger one number is than another.

Yesterday the exponent laws finished — numbers and letters.

Today they go to work on twenty-digit numbers.

Warm-Up

On your own. Four minutes. No notes.

1. Rewrite with a positive exponent: $4x^{-3}$
2. Find the value: $(10^2)^3$
3. Find the value: $\sqrt{121}$
4. In 4,500,000, which place is the 4 in?

Warm-Up

Answers.

1. Rewrite with a positive exponent: $4x^{-3}$ **$4/x^3$**
2. Find the value: $(10^2)^3$ **$10^6 = 1,000,000$**
3. Find the value: $\sqrt{121}$ **11**
4. In 4,500,000, which place is the 4 in? **millions**

Notes

Copy the table.

I. Powers of ten name the places

10^1	10^2	10^3	10^4	10^5	10^6
10	100	1,000	10,000	100,000	1,000,000

The exponent counts the zeros — and names the place of the 1.

Notes

II. Scientific notation

$$a \times 10^n \text{ with } 1 \leq a < 10$$

$$2,000,000 = 2 \times 10^6 \qquad 26,000,000 = 2.6 \times 10^7$$

The coefficient is at least 1 and less than 10: one nonzero digit before the decimal point.

The exponent is how many places the decimal point moves: 26,000,000. → 2.6, seven places, so 10^7 .

Notes

III. How many times larger?

$$\frac{6.4 \times 10^7}{3.2 \times 10^7} = 2$$

$$\frac{2 \times 10^5}{2 \times 10^2} = 10^3 = 1,000$$

$$\frac{8.4 \times 10^6}{2.1 \times 10^4} = 4 \times 10^2 = 400$$

Divide the coefficients. Divide the powers of 10. Multiply the two results.

On a screen, 8.4E6 means 8.4×10^6 .

Example 1

Standard form to scientific notation.

In 2018, funding for education in Florida totaled 30,100,000,000 dollars.

Express this value in scientific notation.

Example 1

Worked.

$$30,100,000,000 = 3.01 \times 10^{10} \quad \text{the 3 is in the ten-billions place}$$

$$3.01 \times 10,000,000,000 = 30,100,000,000 \quad \text{check}$$

Answer: 3.01×10^{10}

Your Turn

Same steps, your numbers. Boards up when done.

A painting sold for 110,500,000 dollars. Write this value in scientific notation.

Your Turn

Answer.

A painting sold for 110,500,000 dollars. Write this value in scientific notation.

the 1 is in the hundred-millions place

Answer: 1.105×10^8

Example 2

How many times larger?

How many times larger is 8.1×10^8 than 5.4×10^7 ?

Example 2

Worked.

$$\frac{8.1}{5.4} = 1.5 \quad \text{coefficients}$$

$$\frac{10^8}{10^7} = 10^1 = 10 \quad \text{powers of 10: subtract the exponents}$$

$$1.5 \times 10 = 15 \quad \text{multiply the two results}$$

Answer: 15 times larger

Your Turn

Same steps, your numbers. Boards up when done.

Everglades alligators: 2.0×10^5

Florida alligators: 1.5×10^6

About how many times larger is the Florida number?

Your Turn

Answer.

Everglades alligators: 2.0×10^5

Florida alligators: 1.5×10^6

About how many times larger is the Florida number?

$$0.75 \times 10 = 7.5$$

Answer: about 7.5 times larger

Whiteboards • Question 1 of 9

Boards up on three.

Write 4,000,000 in scientific notation.

Answer it.

Which place is the 4 in?

Whiteboards • Question 1 of 9

Answer.

Write 4,000,000 in scientific notation.

4 millions

Answer: 4×10^6

Whiteboards • Question 2 of 9

Boards up on three.

Write 7.2×10^4 in standard form.

Answer it.

Move the point four places.

Whiteboards · Question 2 of 9

Answer.

Write 7.2×10^4 in standard form.

$$7.2 \times 10,000$$

Answer: 72,000

Whiteboards • Question 3 of 9

Boards up on three.

A stadium's renovation used 356,000 bolts.

The stadium seats 71,000 people.

Write the number of bolts in scientific notation.

Answer it.

Whiteboards • Question 3 of 9

Answer.

A stadium's renovation used 356,000 bolts.

The stadium seats 71,000 people.

Write the number of bolts in scientific notation.

3.56 hundred-thousands

Answer: 3.56×10^5

Whiteboards • Question 4 of 9

Boards up on three.

Which is greater, 9.5×10^5 or 1.2×10^6 ?

Answer it.

Look at the powers first.

Whiteboards • Question 4 of 9

Answer.

Which is greater, 9.5×10^5 or 1.2×10^6 ?

950,000 versus 1,200,000

Answer: 1.2×10^6

Whiteboards • Question 5 of 9

Boards up on three.

Which is 52,000,000 written in scientific notation?

A. 5.2×10^8

B. 5.2×10^7

C. 52×10^6

D. 5.2×10^6

Answer it.

Whiteboards • Question 5 of 9

Answer.

Which is 52,000,000 written in scientific notation?

A. 5.2×10^8

B. 5.2×10^7

C. 52×10^6

D. 5.2×10^6

Answer: B — 5.2×10^7

Whiteboards • Question 6 of 9

Boards up on three.

How many times larger is 6×10^8 than 6×10^5 ?

A. 1.6

B. 3

C. 1,000

D. 300

Answer it.

Whiteboards • Question 6 of 9

Answer.

How many times larger is 6×10^8 than 6×10^5 ?

A. 1.6

B. 3

C. 1,000

D. 300

Answer: C — 1,000

Whiteboards • Question 7 of 9

Boards up on three.

How many times larger is 8.4×10^6 than 2.1×10^4 ?

Answer it.

Coefficients, then powers, then multiply.

Whiteboards • Question 7 of 9

Answer.

How many times larger is 8.4×10^6 than 2.1×10^4 ?

$$4 \times 10^2$$

Answer: 400

Whiteboards • Question 8 of 9

Boards up on three.

A computer displays a number as $3.5\text{E}9$.

Write the number in standard form.

Answer it.

E is not an error.

Whiteboards • Question 8 of 9

Answer.

A computer displays a number as 3.5E9.

Write the number in standard form.

$$3.5 \times 10^9$$

Answer: 3,500,000,000

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

Marcus writes 2,000,000 as 20×10^5 .

Is his expression equal to 2,000,000? Is it scientific notation? Explain.

Write your answer in sentences.

Use the definition.

Whiteboards · Question 9 of 9

Answer.

Marcus writes 2,000,000 as 20×10^5 .

Is his expression equal to 2,000,000? Is it scientific notation? Explain.

equal, yes — scientific notation, no

Answer: Equal, yes. Scientific notation, no: 20 is not less than 10. 2×10^6 .

Independent Set

Six questions. On your own, in silence.

1. Six questions. Work down the page.
2. Show the step that does the work, not just the answer.
3. Silence until the six minutes are up.

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Convert between standard and scientific notation: large numbers — 2AL

Compare large numbers — WEZ

Powers of ten — LJQ

Writing Small Numbers in Scientific Notation

MA.8.NSO.1.4

I can write very small numbers in scientific notation, read them back, and find how many times larger or smaller one number is than another.

Yesterday the exponents on the 10 were positive.

Today they are negative — and the numbers get very small.

Warm-Up

On your own. Four minutes. No notes.

1. Write 4,300,000 in scientific notation.
2. Write 10^{-3} as a fraction.
3. $\sqrt{95}$ is between which two whole numbers?
4. Write 0.007 as a fraction.

Warm-Up

Answers.

1. Write 4,300,000 in scientific notation. **4.3×10^6**
2. Write 10^{-3} as a fraction. **$\frac{1}{1,000}$**
3. $\sqrt{95}$ is between which two whole numbers? **9 and 10**
4. Write 0.007 as a fraction. **$\frac{7}{1,000}$**

Notes

Copy the table.

I. Negative powers of ten name the decimal places

10^0	10^{-1}	10^{-2}	10^{-3}	10^{-4}	10^{-5}
1	0.1	0.01	0.001	0.0001	0.00001

$$10^{-n} = \frac{1}{10^n} \qquad 10^{-3} = \frac{1}{1,000} = 0.001$$

A negative exponent puts the 1 in that decimal place: 10^{-4} is a 1 in the fourth place.

Notes

II. Small numbers in scientific notation

$$0.000007 = 7 \times 10^{-6} \qquad 0.0000672 = 6.72 \times 10^{-5}$$

$$3.4 \times 10^{-9} = 0.0000000034$$

The coefficient is still at least 1 and less than 10. The exponent is negative for a number less than 1.

The exponent is how many places the point moves: $0.0000672 \rightarrow 6.72$ is five places right, so 10^{-5} .

Notes

III. How many times larger — or smaller?

$$\frac{7.2 \times 10^{-4}}{1.8 \times 10^{-7}} = 4 \times 10^3 = 4,000$$

$$\frac{3 \times 10^8}{3 \times 10^{-4}} = 10^{12}$$

Divide the coefficients. Divide the powers of 10 — subtract the exponents. Multiply the results.

Which is bigger, 10^{-5} or 10^{-8} ? The one with the larger exponent: $-5 > -8$, so 10^{-5} .

Example 1

A quantity in words, to scientific notation.

The density of oxygen is 1,332 millionths of a gram per cubic centimeter.

Write this value as a decimal, then in scientific notation.

Example 1

Worked.

$$\frac{1,332}{1,000,000} = 0.001332 \quad \text{1,332 millionths, as a decimal}$$

$$0.001332 = 1.332 \times 10^{-3} \quad \text{the 1 is in the thousandths place}$$

Answer: $0.001332 = 1.332 \times 10^{-3}$

Your Turn

Same steps, your numbers. Boards up when done.

A human hair is about 67 hundred-thousandths of an inch thick.

Write this value in scientific notation.

Your Turn

Answer.

A human hair is about 67 hundred-thousandths of an inch thick.

Write this value in scientific notation.

0.00067 — the 6 is in the ten-thousandths place

Answer: 6.7×10^{-4}

Example 2

How many times larger?

How many times larger is 7.2×10^{-4} than 1.8×10^{-7} ?

Example 2

Worked.

$$\frac{7.2}{1.8} = 4 \quad \text{coefficients}$$

$$\frac{10^{-4}}{10^{-7}} = 10^{-4 - (-7)} = 10^3 \quad \text{powers of 10: } -4 - (-7) = 3$$

$$4 \times 10^3 = 4,000 \quad \text{multiply the two results}$$

Answer: 4,000 times larger

Your Turn

Same steps, your numbers. Boards up when done.

How many times larger is 5.2×10^{-3} than 2.6×10^{-8} ?

Your Turn

Answer.

How many times larger is 5.2×10^{-3} than 2.6×10^{-8} ?

2×10^5

Answer: 200,000 times larger

Whiteboards • Question 1 of 9

Boards up on three.

$$10^{-4}$$

Answer it.

Write it as a decimal.

Whiteboards • Question 1 of 9

Answer.

$$10^{-4}$$

a 1 in the fourth decimal place

Answer: 0.0001

Whiteboards • Question 2 of 9

Boards up on three.

Write 0.003 in scientific notation.

Answer it.

Which place is the 3 in?

Whiteboards · Question 2 of 9

Answer.

Write 0.003 in scientific notation.

3 thousandths

Answer: 3×10^{-3}

Whiteboards • Question 3 of 9

Boards up on three.

Write 2.5×10^{-4} in standard form.

Answer it.

Move the point four places left.

Whiteboards • Question 3 of 9

Answer.

Write 2.5×10^{-4} in standard form.

the 2 lands in the ten-thousandths place

Answer: 0.00025

Whiteboards • Question 4 of 9

Boards up on three.

A machine's cutting tolerance is 0.0000988 inch.

The part it cuts is 4 inches long.

Write the tolerance in scientific notation.

Answer it.

Whiteboards • Question 4 of 9

Answer.

A machine's cutting tolerance is 0.0000988 inch.

The part it cuts is 4 inches long.

Write the tolerance in scientific notation.

the 9 is in the hundred-thousandths place

Answer: 9.88×10^{-5}

Whiteboards • Question 5 of 9

Boards up on three.

Which is 0.00081 written in scientific notation?

A. 8.1×10^{-3}

B. 8.1×10^4

C. 81×10^{-5}

D. 8.1×10^{-4}

Answer it.

Whiteboards • Question 5 of 9

Answer.

Which is 0.00081 written in scientific notation?

A. 8.1×10^{-3}

B. 8.1×10^4

C. 81×10^{-5}

D. 8.1×10^{-4}

Answer: D — 8.1×10^{-4}

Whiteboards • Question 6 of 9

Boards up on three.

How many times larger is 6×10^{-3} than 2×10^{-5} ?

A. 300

B. 30

C. 3

D. 0.03

Answer it.

Whiteboards • Question 6 of 9

Answer.

How many times larger is 6×10^{-3} than 2×10^{-5} ?

A. 300

B. 30

C. 3

D. 0.03

Answer: A — 300

Whiteboards • Question 7 of 9

Boards up on three.

Which is greater, 3.9×10^{-5} or 7.4×10^{-8} ?

Answer it.

Compare the exponents first.

Whiteboards • Question 7 of 9

Answer.

Which is greater, 3.9×10^{-5} or 7.4×10^{-8} ?

$-5 > -8$, so 10^{-5} is the larger power

Answer: 3.9×10^{-5}

Whiteboards • Question 8 of 9

Boards up on three.

A computer displays a number as $4.2\text{E}-6$.

Write the number in standard form.

Answer it.

E-6 means $\times 10^{-6}$.

Whiteboards • Question 8 of 9

Answer.

A computer displays a number as $4.2\text{E}-6$.

Write the number in standard form.

$$4.2 \times 10^{-6}$$

Answer: 0.0000042

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

Fishing line A is 2×10^{-2} inch thick.

Fishing line B is 6×10^{-2} inch thick.

Which is thicker, and how many times? Name a thickness between them.

Write your answer in sentences.

Same power of 10 — compare the coefficients.

Whiteboards • Question 9 of 9

Answer.

Fishing line A is 2×10^{-2} inch thick.

Fishing line B is 6×10^{-2} inch thick.

Which is thicker, and how many times? Name a thickness between them.

0.02 in and 0.06 in

Answer: B; 3 times; e.g. 4×10^{-2} in.

Independent Set

Six questions. On your own, in silence.

1. Six questions. Work down the page.
2. Show the step that does the work, not just the answer.
3. Silence until the six minutes are up.

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Convert between standard and scientific notation: small numbers — 85M

Compare small numbers — KDR

Powers of ten with negative exponents — SKD